3.14 Filtered Rings and Completions

Both DVRs and complete DVRs sit naturally in the category of filtered rings developed in this subsection: a DVR such as $A_{(p)}$ is filtered by the powers of its maximal ideal, and a complete DVR such as $\mathbb{Z}_p$ or $k[[X]]$ is exactly such a filtered ring whose canonical map to its own completion happens to be an isomorphism (3.14.20). Working at this level of generality – filtered rings and continuous homomorphisms between them, rather than DVRs and local homomorphisms specifically – gives a category closed under more operations than DVRs themselves support: for instance, an arbitrary product of filtered rings is again filtered, with completion computed factorwise (3.14.22), even though a product of DVRs is essentially never itself a DVR. This is the kind of fact the filtered-ring framework makes easy to state: the completion of a semi-local Dedekind domain (such as the integral closure $\mathcal{O}_L$ of a DVR in a finite extension) agrees with the product of the completions at its finitely many primes, by the Chinese Remainder Theorem factorwise – though the trace/residue theory developed later avoids needing this, working directly with Laurent series instead (see the Extensions of Power Series Rings section).

Filtered rings, at this level of generality, are the algebraic core of the **admissible rings** of [Gro60] (0_{I}, §7) (linearly topologized rings with a fundamental system of open ideals, not necessarily powers of a single ideal).

Definition 3.14.1 (Filtered Ring)
*A **filtered ring** $(A, \mathfrak{a}_\bullet)$ is a ring A together with a decreasing chain of ideals $\mathfrak{a}_\bullet = (\mathfrak{a}_n)_{n\in\mathbb{Z}}$ of A, that is $\mathfrak{a}_{n+1} \subseteq \mathfrak{a}_n$ for every $n \in \mathbb{Z}$, with $\mathfrak{a}_n = A$ for every $n < 0$. We say $(A, \mathfrak{a}_\bullet)$ is **separated** if $\bigcap_n \mathfrak{a}_n = 0$.*

Definition 3.14.2 (Filtered Module)
*Let A be a ring and M an A-module. A **filtration** of M (as an A-module) is a decreasing chain $\mathcal{M}_\bullet = (\mathcal{M}_n)_{n\in\mathbb{Z}}$ of A-submodules of M, that is $\mathcal{M}_{n+1} \subseteq \mathcal{M}_n$ for every $n \in \mathbb{Z}$. We call $(M, \mathcal{M}_\bullet)$ a **filtered A-module**, and say it is **separated** if $\bigcap_n \mathcal{M}_n = 0$, and **exhaustive** if $\bigcup_n \mathcal{M}_n = M$.*

Every positively filtered A-module (3.14.3) is exhaustive, trivially ($\mathcal{M}_n = M$ already for $n < 0$); the fractional-ideal filtration of $K = \mathrm{Frac}(A)$, A a DVR, is a genuinely exhaustive example that is not positively filtered.

*When $(A, \mathfrak{a}_\bullet)$ is itself a filtered ring, every A-module M carries a canonical filtration $\mathfrak{a}_\bullet M := (\mathfrak{a}_n M)_{n\in\mathbb{Z}}$, the **induced filtration** (with $\mathfrak{a}_n M = M$ for $n < 0$, as $\mathfrak{a}_n = A$ there; so $(M, \mathfrak{a}_\bullet M)$ is positively filtered, hence exhaustive).*

Definition 3.14.3 (Positively Filtered)
*A filtered ring (3.14.1), or more generally a filtered A-module $(M, \mathcal{M}_\bullet)$ (3.14.2), is **positively filtered** if $\mathcal{M}_n$ equals the whole module for every $n < 0$. Taking $M := A$, $\mathcal{M}_n := \mathfrak{a}_n$ exhibits every filtered ring as a positively filtered A-module over itself.*

Definition 3.14.4 ($\mathfrak{a}$-adic Filtration)
*Let A be a ring and $\mathfrak{a} \triangleleft A$ an ideal. The **$\mathfrak{a}$-adic filtration** is $\mathfrak{a}_n := \mathfrak{a}^{n+1}$ for $n \geq 0$ and $\mathfrak{a}_n := A$ for $n < 0$, a filtration of A (3.14.1).*

Definition 3.14.5 (Continuous Homomorphism)
*Let $(A, \mathfrak{a}_\bullet)$, $(B, \mathfrak{b}_\bullet)$ be filtered rings. A ring homomorphism $\phi : A \to B$ is **continuous** if, for every $m \geq 0$, there is $n \geq 0$ with $\phi(\mathfrak{a}_n) \subseteq \mathfrak{b}_m$.*

Remark 3.14.6
Continuity is exactly continuity of ϕ at 0 in the usual metric-space sense, for the metrics $d_A(x,y) := 2^{-\sup\{n : x-y\in\mathfrak{a}_n\}}$ on A and $d_B(x,y) := 2^{-\sup\{n : x-y\in\mathfrak{b}_n\}}$ on B (with $\sup\varnothing := -1$, so d_A, d_B vanish exactly on equal pairs): a ball of radius 2^{-m} about $0 \in B$ is exactly $\mathfrak{b}_{m+1}$, so "$\phi(\mathfrak{a}_{n+1}) \subseteq \mathfrak{b}_{m+1}$ for some n, for every m" is exactly continuity of ϕ at 0. As d_A, d_B are translation-invariant ($d(x,y) = d(x-y, 0)$, immediate as $\mathfrak{a}_n, \mathfrak{b}_n$ are subgroups), this already gives continuity of ϕ at every point of A: given $x_0 \in A$ and $\varepsilon > 0$, pick m with $2^{-m} \leq \varepsilon$ and n with $\phi(\mathfrak{a}_{n+1}) \subseteq \mathfrak{b}_{m+1}$; then $d_A(x, x_0) < 2^{-n}$ gives $x - x_0 \in \mathfrak{a}_{n+1}$, so $\phi(x) - \phi(x_0) = \phi(x - x_0) \in \mathfrak{b}_{m+1}$, that is $d_B(\phi(x), \phi(x_0)) \leq 2^{-m} \leq \varepsilon$. (This is the same translation-invariance trick used for the identity map in (3.14.13), which compares two filtrations of a single ring this way.)

For the remainder of this subsection, fix filtered rings $(A, \mathfrak{a}_\bullet)$, $(A', \mathfrak{a}'_\bullet)$, a ring homomorphism $\rho : A \to A'$, and filtered A-, A'-modules $(M, \mathcal{M}_\bullet)$, $(M', \mathcal{M}'_\bullet)$ (3.14.2): this notation is reused without restatement in every definition and proposition below phrased in terms of a map $\psi : M \to M'$.

Definition 3.14.7 (Continuous Homomorphism, Module Case)
*An additive map $\psi : M \to M'$ is a **ρ-linear map** (or **ρ-homomorphism**) if $\psi(a \cdot x) = \rho(a) \cdot \psi(x)$ for every $a \in A$, $x \in M$, and is **continuous** if, for every $m \in \mathbb{Z}$, there is $n \in \mathbb{Z}$ with $\psi(\mathcal{M}_n) \subseteq \mathcal{M}'_m$.*

© David Rufino CC BY-NC-ND 4.0

When $A = A'$ and $\rho = \mathrm{id}_A$, a continuous ρ-homomorphism is exactly an ordinary continuous A-module homomorphism.

Definition 3.14.8 (Dense Homomorphism)
*Let $(A, \mathfrak{a}_\bullet)$, $(B, \mathfrak{b}_\bullet)$ be filtered rings. A ring homomorphism $\phi : A \to B$ is **dense** if $B = \phi(A) + \mathfrak{b}_n$ for every $n \geq 0$.*

Definition 3.14.9 (Dense Homomorphism, Module Case)
*A ρ-homomorphism $\psi : M \to M'$ (3.14.7) is **dense** if $M' = \psi(M) + \mathcal{M}'_n$ for every $n \geq 0$. (Taking $M := A$, $M' := A'$ as filtered A-, A'-modules over themselves and $\psi := \rho$ recovers (3.14.8).)*

Definition 3.14.10 (Taut Homomorphism)
*Let $(A, \mathfrak{a}_\bullet)$, $(B, \mathfrak{b}_\bullet)$ be filtered rings and $\phi : A \to B$ a ring homomorphism with $\phi(\mathfrak{a}_n) \subseteq \mathfrak{b}_n$ for every $n \geq 0$. Then ϕ is **taut** if there is a strictly increasing sequence $n_0 < n_1 < n_2 < \cdots$ in $\mathbb{N}$ such that*

$$B = \phi(A) + \mathfrak{b}_{n_0} \qquad \text{and} \qquad \mathfrak{b}_{n_N} = \phi\big(\mathfrak{a}_{n_N}\big) + \mathfrak{b}_{n_{N+1}} \text{ for every } N \geq 0.$$

Definition 3.14.11 (Taut Homomorphism, Module Case)
*A ρ-homomorphism $\psi : M \to M'$ with $\psi(\mathcal{M}_n) \subseteq \mathcal{M}'_n$ for every $n \geq 0$ is **taut** if there is a strictly increasing sequence $n_0 < n_1 < n_2 < \cdots$ in $\mathbb{N}$ such that*

$$M' = \psi(M) + \mathcal{M}'_{n_0} \qquad \text{and} \qquad \mathcal{M}'_{n_N} = \psi\big(\mathcal{M}_{n_N}\big) + \mathcal{M}'_{n_{N+1}} \text{ for every } N \geq 0.$$

(Taking $M := A$, $M' := A'$ as filtered A-, A'-modules over themselves and $\psi := \rho$ recovers (3.14.10).)

Remark 3.14.12
*The name is borrowed, by analogy, from the **taut** ring maps of linearly topologized rings in [Stacks] (Tag 0AMX) – itself flagged there as nonstandard terminology – where an analogous (though not identical) condition, combined with density and completeness, is used to conclude surjectivity via an open mapping theorem ([Stacks] Tag 0APT) rather than the direct successive-approximation argument used here.*

Proposition 3.14.13 (Refinement Criterion)
Let $\mathfrak{a}_\bullet$, $\mathfrak{a}'_\bullet$ be filtrations of a ring A (3.14.1), with metrics $d_1(x,y) := 2^{-\sup\{n : x-y \in \mathfrak{a}_n\}}$ and $d_2(x,y) := 2^{-\sup\{n : x-y \in \mathfrak{a}'_n\}}$ on A ($\sup \varnothing := -1$). The following are equivalent:

a) $1_A : (A, \mathfrak{a}_\bullet) \to (A, \mathfrak{a}'_\bullet)$ is continuous (3.14.5)
*b) $\mathfrak{a}_\bullet$ is **cofinal** in $\mathfrak{a}'_\bullet$: for every $m \geq 0$ there is $n \geq 0$ with $\mathfrak{a}_n \subseteq \mathfrak{a}'_m$;*
c) $1_A : (A, d_1) \to (A, d_2)$ is continuous at 0 in the metric-space sense;
d) the topology on A induced by d_2 is weaker than the one induced by d_1: every d_2-open subset of A is d_1-open.

Proof. Drafted by Claude Sonnet 5, reviewed by David Rufino.

a) $\iff$ b) is (3.14.5) applied to $\phi := 1_A$.

b) $\implies$ c): given $\varepsilon > 0$, pick $m \geq 0$ with $2^{-m} \leq \varepsilon$; by b) at $m+1$, pick $n \geq 0$ with $\mathfrak{a}_n \subseteq \mathfrak{a}'_{m+1}$, and set $\delta := 2^{-(n-1)}$ (any $\delta > 0$ if $n = 0$). If $d_1(x,0) < \delta$ then $x \in \mathfrak{a}_n \subseteq \mathfrak{a}'_{m+1}$, so $d_2(x,0) \leq 2^{-m} \leq \varepsilon$.

c) $\implies$ b): given $m \geq 0$, apply c) with $\varepsilon := 2^{-m}$ to get $\delta > 0$ with $d_1(x,0) < \delta \implies d_2(x,0) \leq 2^{-m}$; picking $n \geq 0$ with $2^{-(n-1)} < \delta$ (any $n \geq 1$ if δ exceeds every such bound) gives $\mathfrak{a}_n \subseteq \{x : d_1(x,0) < \delta\} \subseteq \{x : d_2(x,0) \leq 2^{-m}\} = \mathfrak{a}'_m$.

c) $\implies$ d): both d_1, d_2 are translation-invariant ($d_i(x+z, y+z) = d_i(x,y)$, immediate as $(x+z)-(y+z) = x-y$), so continuity of $1_A : (A, d_1) \to (A, d_2)$ at 0 gives continuity at every point $x_0 \in A$: given $\varepsilon > 0$, take $\delta > 0$ as in c); if $d_1(x, x_0) < \delta$ then $d_1(x - x_0, 0) < \delta$, so $d_2(x - x_0, 0) < \varepsilon$, that is $d_2(x, x_0) < \varepsilon$. Continuity of 1_A at every point is exactly d) (the preimage, under the identity map, of a d_2-open set is itself, and it is d_1-open).

d) $\implies$ c): a d_2-ball around 0 is d_2-open, hence d_1-open by d), hence contains a d_1-ball around 0 (as 0 lies in it); this is exactly continuity of 1_A at 0. ∎

Proposition 3.14.14 (Equivalence Criterion)
Let $\mathfrak{a}_\bullet$, $\mathfrak{a}'_\bullet$ be filtrations of a ring A (3.14.1), with metrics d_1, d_2 as in (3.14.13). The following are equivalent:

*a) $1_A : (A, \mathfrak{a}_\bullet) \to (A, \mathfrak{a}'_\bullet)$ is **bicontinuous**, that is 1_A is continuous both as a map $(A, \mathfrak{a}_\bullet) \to (A, \mathfrak{a}'_\bullet)$ and as a map $(A, \mathfrak{a}'_\bullet) \to (A, \mathfrak{a}_\bullet)$ (3.14.5)*
b) $\mathfrak{a}_\bullet$ and $\mathfrak{a}'_\bullet$ are cofinal in each other (3.14.13)
*c) d_1 and d_2 are **equivalent** metrics on A: 1_A is bicontinuous at 0 as a map $(A, d_1) \to (A, d_2)$;*
d) d_1 and d_2 induce the same *topology on A.*

© David Rufino CC BY-NC-ND 4.0

Proof. Drafted by Claude Sonnet 5, reviewed by David Rufino.

Applying (3.14.13) to the pair $(\mathfrak{a}_\bullet, \mathfrak{a}'_\bullet)$ gives the equivalence of "$\mathfrak{a}_\bullet$ cofinal in $\mathfrak{a}'_\bullet$", "$1_A : (A, \mathfrak{a}_\bullet) \to (A, \mathfrak{a}'_\bullet)$ continuous", "$1_A : (A, d_1) \to (A, d_2)$ continuous at 0", and "every d_2-open set is d_1-open"; applying it to the reversed pair $(\mathfrak{a}'_\bullet, \mathfrak{a}_\bullet)$ gives the equivalence of the four corresponding statements with the roles of $1, 2$ exchanged. Conjoining each pair of equivalences (an "if and only if" together with its reverse) gives exactly a) $\iff$ b) $\iff$ c) $\iff$ d), the last as "every d_2-open set is d_1-open, and every d_1-open set is d_2-open" is exactly the two topologies having the same open sets. ∎

Lemma 3.14.15 (Continuity Criterion for Ideal-Power Filtrations)
Let A, B be rings with the $\mathfrak{a}$-adic and $\mathfrak{b}$-adic filtrations (3.14.4), for an ideal $\mathfrak{a} \lhd A$ and a radical *ideal $\mathfrak{b} \lhd B$ (3.3.35). Then a ring homomorphism $\phi : A \to B$ is continuous (3.14.5) if and only if $\phi(\mathfrak{a}) \subseteq \mathfrak{b}$.*

Proof. Drafted by Claude Sonnet 5, reviewed by David Rufino.

If $\phi(\mathfrak{a}) \subseteq \mathfrak{b}$: for every $k \geq 1$, $\phi\big(\mathfrak{a}^k\big)$ is generated (as an ideal of B) by products of k elements of $\phi(\mathfrak{a}) \subseteq \mathfrak{b}$, hence $\phi\big(\mathfrak{a}^k\big) \subseteq \mathfrak{b}^k$; so, given m, taking $n := m$ gives $\phi(\mathfrak{a}_n) = \phi\big(\mathfrak{a}^{n+1}\big) \subseteq \mathfrak{b}^{m+1} = \mathfrak{b}_m$: ϕ is continuous.

Conversely, suppose ϕ is continuous; taking $m := 0$ gives some n with $\phi\big(\mathfrak{a}^{n+1}\big) \subseteq \mathfrak{b}$. If $\phi(\mathfrak{a})\neg \subseteq \mathfrak{b}$, pick $a \in \mathfrak{a}$ with $\phi(a) \notin \mathfrak{b}$; as $\mathfrak{b}$ is radical, its defining property $x^{n+1} \in \mathfrak{b} \implies x \in \mathfrak{b}$ contraposes to $\phi(a) \notin \mathfrak{b} \implies \phi(a)^{n+1} \notin \mathfrak{b}$. But $\phi(a)^{n+1} = \phi(a^{n+1}) \in \phi\big(\mathfrak{a}^{n+1}\big) \subseteq \mathfrak{b}$, as $a^{n+1} \in \mathfrak{a}^{n+1}$: a contradiction. So $\phi(\mathfrak{a}) \subseteq \mathfrak{b}$. ∎

Definition 3.14.16 (Completion of a Filtered Ring)
Let $(A, \mathfrak{a}_\bullet)$ be a filtered ring. For $n \geq m \geq 0$, as $\mathfrak{a}_n \subseteq \mathfrak{a}_m$, (3.3.55) gives a unique ring homomorphism $\rho_{n,m} : A/\mathfrak{a}_n \to A/\mathfrak{a}_m$ making

$$\begin{array}{ccc} A & \twoheadrightarrow & A/\mathfrak{a}_n \\ & \searrow & \downarrow{\scriptstyle \rho_{n,m}} \\ & & A/\mathfrak{a}_m \end{array}$$

commute; in particular it is surjective. These satisfy $\rho_{n,n} = 1_{A/\mathfrak{a}_n}$ and, for $n \geq k \geq m \geq 0$,

$$\rho_{k,m} \circ \rho_{n,k} = \rho_{n,m}$$

(both sides make the outer triangle $A \twoheadrightarrow A/\mathfrak{a}_n \to A/\mathfrak{a}_m$ commute, so agree by the uniqueness clause of (3.3.55)). Specialising to $(n, m) := (n+1, n)$ gives reduction maps $\rho_{n+1,n} : A/\mathfrak{a}_{n+1} \twoheadrightarrow A/\mathfrak{a}_n$, which form a tower (3.3.153) the ***completion*** *of $(A, \mathfrak{a}_\bullet)$ is*

$$\hat{A} := \varprojlim_n A/\mathfrak{a}_n,$$

with projections $\lambda_n : \hat{A} \to A/\mathfrak{a}_n$ (3.3.153) and the canonical ring homomorphism

$$\iota : A \to \hat{A}, \quad a \mapsto (a \bmod \mathfrak{a}_n)_{n \geq 0}.$$

Consequently, every $x = (a_n)_{n \geq 0} \in \hat{A}$ satisfies the ***repeated compatibility*** *$\rho_{n,m}(a_n) = a_m$ for all $n \geq m \geq 0$, not just consecutive $n = m + 1$: by induction on $n - m$, composing the defining consecutive relations $\rho_{k,k-1}(a_k) = a_{k-1}$ for $m < k \leq n$.*

Moreover $\ker \lambda_\bullet := (\ker \lambda_n)_{n \geq 0}$ is itself a decreasing chain of ideals of $\hat{A}$ – for $x = (\overline{a}_n)_n$, $\lambda_n(x) = 0$ forces $\lambda_{n-1}(x) = 0$ too (the further reduction of $\overline{a}_n = 0$ is 0) – so $\big(\hat{A}, \ker \lambda_\bullet\big)$ is again a filtered ring (3.14.1).

Definition 3.14.17 (Completion of a Filtered Module)
Let A be a ring and $(M, \mathcal{M}_\bullet)$ a filtered A-module (3.14.2). For $n \geq m \geq 0$, as $\mathcal{M}_n \subseteq \mathcal{M}_m$, (3.3.100) gives a unique A-module homomorphism making

$$\begin{array}{ccc} M & \twoheadrightarrow & M/\mathcal{M}_n \\ & \searrow & \downarrow \\ & & M/\mathcal{M}_m \end{array}$$

commute; in particular it is surjective. Specialising to $(n, m) := (n+1, n)$ gives reduction maps $M/\mathcal{M}_{n+1} \twoheadrightarrow M/\mathcal{M}_n$, which form a tower of A-modules (3.3.153) the ***completion*** *of $(M, \mathcal{M}_\bullet)$ is the A-module*

$$\widehat{M} := \varprojlim_n M/\mathcal{M}_n,$$

© David Rufino CC BY-NC-ND 4.0

with projections $\lambda_n : \widehat{M} \to M/\mathcal{M}_n$ *(3.3.153) and the canonical* A*-module homomorphism*

$$\iota : M \to \widehat{M}, \quad v \mapsto (v \bmod \mathcal{M}_n)_{n\geq 0}.$$

Moreover $\ker \lambda_\bullet := (\ker \lambda_n)_{n\geq 0}$ *is itself a decreasing chain of* A*-submodules of* $\widehat{M}$ *– for* $x = (\overline{v}_n)_n$*,* $\lambda_n(x) = 0$ *forces* $\lambda_{n-1}(x) = 0$ *too (the further reduction of* $\overline{v}_n = 0$ *is* 0*).*

Taking $M := A$ *as a filtered* A*-algebra over itself (3.14.3),* $\mathcal{M}_n = \mathfrak{a}_n$ *is an ideal of* $M = A$*, so each* $M/\mathcal{M}_n$ *is also a ring and* ι *a ring homomorphism ((3.3.55) giving the same reduction maps as ring homomorphisms): this recovers exactly the completion of the filtered ring* $(A, \mathfrak{a}_\bullet)$ *(3.14.16).*

Corollary 3.14.18 (Completion via a Cofinal Sequence)
Let $(M, \mathcal{M}_\bullet)$ *be a filtered* A*-module, with completion* $\widehat{M}$*,* λ_n *(3.14.17), and* $n_0 < n_1 < n_2 < \cdots$ *a strictly increasing sequence in* $\mathbb{N}$*, with projections* λ'_N *of* $\varprojlim_N M/\mathcal{M}_{n_N}$*. Then*

$$\theta : \widehat{M} \xrightarrow{\sim} \varprojlim_N M/\mathcal{M}_{n_N}, \quad \theta\Big((\overline{v}_n)_{n\geq 0}\Big) = \big(\overline{v}_{n_N}\big)_{N\geq 0},$$

is a bicontinuous A*-module isomorphism, with* $\lambda'_N \circ \theta = \lambda_{n_N}$ *for every* $N \geq 0$*. (When* $M := A$*, this is a bicontinuous ring isomorphism.)*

Proof. Drafted by Claude Sonnet 5, reviewed by David Rufino.

That θ is an A-module isomorphism with $\lambda'_N \circ \theta = \lambda_{n_N}$ for every $N \geq 0$ is (3.3.157), applied to the tower $(M/\mathcal{M}_n)_{n\geq 0}$ of A-modules (3.14.17) and this sequence. As θ is bijective, this gives $\theta\big(\ker \lambda_{n_N}\big) = \ker \lambda'_N$ for every $N \geq 0$.

Continuity of θ. Given $N \geq 0$, taking $n := n_N$: $\theta(\ker \lambda_n) = \ker \lambda'_N$.

Continuity of θ^{-1}. Given $n \geq 0$: as $(n_N)_N$ is unbounded, choose N with $n_N \geq n$, so $\ker \lambda_{n_N} \subseteq \ker \lambda_n$ ((3.14.17), the filtration $\ker \lambda_\bullet$ decreasing); then $\theta^{-1}(\ker \lambda'_N) = \ker \lambda_{n_N} \subseteq \ker \lambda_n$. ∎

Definition 3.14.19 ($\mathfrak{a}$-adic Completion)
Let A *be a ring and* $\mathfrak{a} \triangleleft A$ *an ideal. The* ***$\mathfrak{a}$-adic completion*** *of* A *is the completion (3.14.16) of* A *for the* $\mathfrak{a}$*-adic filtration (3.14.4):*

$$\hat{A}_\mathfrak{a} := \hat{A} = \varprojlim_n A/\mathfrak{a}^{n+1},$$

with projections $\lambda_n : \hat{A}_\mathfrak{a} \to A/\mathfrak{a}^{n+1}$ *and canonical ring homomorphism*

$$\iota_\mathfrak{a} : A \to \hat{A}_\mathfrak{a}, \quad a \mapsto (a \bmod \mathfrak{a}^{n+1})_{n\geq 0},$$

as in (3.14.16).

Proposition 3.14.20 (Completeness Criterion)
Let $(M, \mathcal{M}_\bullet)$ *be a* separated *filtered* A*-module (3.14.2), and* $\widehat{M}$*,* ι *as in (3.14.17). Then the following are equivalent:*

a) $\iota : M \to \widehat{M}$ *is surjective;*
b) every sequence $(\overline{v}_n)_{n\geq 0}$*,* $\overline{v}_n \in M/\mathcal{M}_n$*, compatible under reduction (*$\overline{v}_n \equiv \overline{v}_{n-1} \bmod \mathcal{M}_{n-1}$ *for* $n \geq 1$*), equals* $(v \bmod \mathcal{M}_n)_n$ *for some* $v \in M$*;*
c) for some strictly increasing sequence $n_0 < n_1 < n_2 < \cdots$ *in* $\mathbb{N}$ *(equivalently, for every such sequence), every sequence* $(\overline{v}_N)_{N\geq 0}$*,* $\overline{v}_N \in M/\mathcal{M}_{n_N}$*, compatible under reduction, equals* $\big(v \bmod \mathcal{M}_{n_N}\big)_N$ *for some* $v \in M$*.*

When these hold, ι *is an* A*-module isomorphism, and we say* $(M, \mathcal{M}_\bullet)$ *is* ***complete****. (When* $M := A$*,* ι *is a ring isomorphism, and we say the filtered ring* $(A, \mathfrak{a}_\bullet)$ *is* ***complete****.)*

Proof. Drafted by Claude Sonnet 5, reviewed by David Rufino.

By (3.3.153), $\widehat{M} = \varprojlim_n M/\mathcal{M}_n$ consists precisely of the compatible sequences $(\overline{v}_n)_{n\geq 0}$ of b); and $\iota(v) = (v \bmod \mathcal{M}_n)_n$ by definition. So b) says exactly that every element of $\widehat{M}$ lies in the image of ι, that is, a).

a) $\iff$ c): fix a strictly increasing sequence $n_0 < n_1 < n_2 < \cdots$ in $\mathbb{N}$. By (3.14.18), $(\overline{v}_n)_{n\geq 0} \mapsto \big(\overline{v}_{n_N}\big)_{N\geq 0}$ is an isomorphism $\widehat{M} \xrightarrow{\sim} \varprojlim_N M/\mathcal{M}_{n_N}$; composing with ι, this sends $v \in M$ to $\big(v \bmod \mathcal{M}_{n_N}\big)_N$, so identifies the image of ι with the image of $v \mapsto \big(v \bmod \mathcal{M}_{n_N}\big)_N$. Hence ι is surjective iff this latter map is, that is, iff c) holds for this particular sequence $(n_N)_N$; as $(n_N)_N$ was an arbitrary strictly increasing sequence, a) is equivalent to c) holding for some such sequence, and also to it holding for every such sequence.

© David Rufino CC BY-NC-ND 4.0

Finally, $\ker\iota=\bigcap_n\mathcal{M}_n=0$ ($(M,\mathcal{M}_\bullet)$ being separated): $\iota(v)=0$ iff $v \bmod \mathcal{M}_n=0$ for every n, that is $v\in\mathcal{M}_n$ for every n. So ι is injective, hence, combined with a), an A-module isomorphism (a ring isomorphism when $M:=A$, as ι is then also a ring homomorphism (3.14.16)). ∎

Proposition 3.14.21 (Completeness Criterion, $\mathfrak{a}$-adic Case)
Let A be a ring, $\mathfrak{a}\triangleleft A$ an ideal with $\bigcap_n\mathfrak{a}^n=0$ (that is, A is separated for the $\mathfrak{a}$-adic filtration), and $\hat{A}_\mathfrak{a}$, $\iota_\mathfrak{a}$ as in (3.14.19). This is the case $B:=A$, $\mathcal{M}_n:=\mathfrak{a}_n:=\mathfrak{a}^{n+1}$ of (3.14.20), giving: the following are equivalent:

- *a) $\iota_\mathfrak{a}:A\to\hat{A}_\mathfrak{a}$ is surjective;*
- *b) every sequence $(\overline{a}_n)_{n\geq0}$, $\overline{a}_n\in A/\mathfrak{a}^{n+1}$, compatible under reduction ($\overline{a}_n\equiv\overline{a}_{n-1} \bmod \mathfrak{a}^n$ for $n\geq1$), equals $\left(a \bmod \mathfrak{a}^{n+1}\right)_n$ for some $a\in A$;*
- *c) for some strictly increasing sequence $n_0<n_1<n_2<\cdots$ in $\mathbb{N}$ (equivalently, for every such sequence), every sequence $(\overline{a}_N)_{N\geq0}$, $\overline{a}_N\in A/\mathfrak{a}^{n_N+1}$, compatible under reduction, equals $\left(a \bmod \mathfrak{a}^{n_N+1}\right)_N$ for some $a\in A$.*

*When these hold, $\iota_\mathfrak{a}$ is an isomorphism, and we say A is **$\mathfrak{a}$-adically complete**.*

Proposition 3.14.22 (Products of Filtered Modules)
Let A be a ring, K a set, and $\left(M_k,\mathcal{M}_{k,\bullet}\right)_{k\in K}$ a family of filtered A-modules (3.14.2), with completions $\widehat{M}_k$, $\lambda_{k,n}$ (3.14.17). Then $\mathcal{M}_\bullet:=\left(\prod_{k\in K}\mathcal{M}_{k,n}\right)_{n\in\mathbb{Z}}$ is a filtration of the A-module $M:=\prod_{k\in K}M_k$, with completion $\widehat{M}$, λ_n. There is a unique A-module homomorphism Ψ making

$$\begin{array}{ccc}\prod_{k\in K}\widehat{M}_k & \xrightarrow{\ \Psi\ } & \widehat{\left(\prod_{k\in K}M_k\right)}\\ \Big\downarrow{\scriptstyle \prod_k\lambda_{k,n}} & & \Big\downarrow{\scriptstyle \lambda_n}\\ \prod_{k\in K}(M_k/\mathcal{M}_{k,n}) & \xleftarrow[\sim]{\ \overline{q}_n\ } & \left(\prod_{k\in K}M_k\right)/\mathcal{M}_n\end{array}$$

commute for every $n\geq0$, where $\overline{q}_n\left(\left(m_k\right)_k \bmod \mathcal{M}_n\right):=\left(m_k \bmod \mathcal{M}_{k,n}\right)_k$; explicitly, writing $\hat{x}_k=\left(\overline{m}_{k,n}\right)_{n\geq0}$ for $\hat{x}_k\in\widehat{M}_k$,

$$\Psi\left(\left(\hat{x}_k\right)_k\right)=\left(\left(m_{k,n}\right)_k \bmod \mathcal{M}_n\right)_{n\geq0}$$

for any representatives $m_{k,n}\in M_k$ of $\overline{m}_{k,n}$; and Ψ is a bicontinuous A-module isomorphism, for the product filtration $\left(\prod_k\ker\lambda_{k,n}\right)_n$ of $\prod_k\widehat{M}_k$ and $\ker\lambda_\bullet$ of $\widehat{M}$. (When K is finite *and each $M_k:=A_k$ is taken to be a filtered ring over itself, Ψ is a bicontinuous ring isomorphism, recovering the completion of the product of filtered rings $\left(A_k,\mathfrak{a}_{k,\bullet}\right)_{k\in K}$ – exhaustiveness of the product filtration, $\bigcup_n\prod_k\mathfrak{a}_{k,n}=\prod_kA_k$, using finiteness of K (taking, for each $(a_k)_k$, $n:=\min_k n_k$ over the finitely many n_k with $a_k\in\mathfrak{a}_{k,n_k}$); for infinite K, exhaustiveness can fail even if every factor is exhaustive, as no single n need work for infinitely many coordinates at once.)*

Proof. Drafted by Claude Sonnet 5, reviewed by David Rufino.

$\mathcal{M}_\bullet$ is a filtration of $\prod_k M_k$. Each $\mathcal{M}_n:=\prod_k\mathcal{M}_{k,n}$ is an A-submodule of $\prod_kM_k$ (a product of submodules, coordinatewise), and $\mathcal{M}_{n+1}\subseteq\mathcal{M}_n$ as $\mathcal{M}_{k,n+1}\subseteq\mathcal{M}_{k,n}$ for every k.

A levelwise isomorphism. For each n, the coordinatewise reduction $q_n:\prod_kM_k\to\prod_k(M_k/\mathcal{M}_{k,n})$, $(m_k)_k\mapsto\left(m_k \bmod \mathcal{M}_{k,n}\right)_k$, is a surjective A-module homomorphism with kernel $\mathcal{M}_n$, so induces an isomorphism $\overline{q}_n:\left(\prod_kM_k\right)/\mathcal{M}_n\xrightarrow{\sim}\prod_k(M_k/\mathcal{M}_{k,n})$ (3.3.101), compatible with reduction as n varies (both sides' reduction maps being coordinatewise reduction). So, by (3.3.155), $\varprojlim\overline{q}:\widehat{\left(\prod_kM_k\right)}\xrightarrow{\sim}\varprojlim_n\left(\prod_k(M_k/\mathcal{M}_{k,n})\right)$ is an isomorphism.

Composing with (3.3.156). That proposition (applied to the towers $\left(M_k/\mathcal{M}_{k,n}\right)_n$ of A-modules) gives an isomorphism $\Phi:\prod_k\widehat{M}_k\xrightarrow{\sim}\varprojlim_n\left(\prod_k(M_k/\mathcal{M}_{k,n})\right)$. Set $\Psi:=\left(\varprojlim\overline{q}\right)^{-1}\circ\Phi$, an isomorphism.

Ψ has the stated property. By the explicit formulas of (3.3.155) and (3.3.156) for the projections of $\varprojlim\overline{q}$ and Φ respectively: for $(\hat{x}_k)_k\in\prod_k\widehat{M}_k$, the n-th coordinate of $\Phi\left(\left(\hat{x}_k\right)_k\right)$ is $\left(\lambda_{k,n}(\hat{x}_k)\right)_k\in\prod_k(M_k/\mathcal{M}_{k,n})$, and this equals $\overline{q}_n\left(\lambda_n\left(\Psi\left(\left(\hat{x}_k\right)_k\right)\right)\right)$ ($\varprojlim\overline{q}$'s n-th coordinate being $\overline{q}_n\circ\lambda_n$); so $\overline{q}_n\left(\lambda_n\left(\Psi\left(\left(\hat{x}_k\right)_k\right)\right)\right)=\left(\lambda_{k,n}(\hat{x}_k)\right)_k=\left(\overline{m}_{k,n}\right)_k$: the square commutes, for every n. Unwinding $\overline{q}_n^{-1}$ (3.3.101), $\lambda_n\left(\Psi\left(\left(\hat{x}_k\right)_k\right)\right)=\left(m_{k,n}\right)_k \bmod \mathcal{M}_n$ for any representatives $m_{k,n}$ of $\overline{m}_{k,n}$, giving the stated explicit formula for Ψ.

© David Rufino CC BY-NC-ND 4.0

Uniqueness. $\left(\widehat{\prod_k M_k}\right) = \varprojlim_n \left(\prod_k M_k\right)/\mathcal{M}_n$ is universal for compatible families of morphisms into the tower $\left(\left(\prod_k M_k\right)/\mathcal{M}_n\right)_n$ (3.3.153): an A-module homomorphism Ψ' making the square commute for every n is exactly a compatible family of morphisms $\prod_k \widehat{M}_k \to \left(\prod_k M_k\right)/\mathcal{M}_n$ (namely $\overline{q}_n^{-1} \circ \prod_k \lambda_{k,n}$, compatible as n varies since the $\lambda_{k,n}$ and $\overline{q}_n$ are), and that universal property gives a unique such morphism: so $\Psi' = \Psi$.

Bicontinuity. Rearranging the commuting square, $\lambda_n \circ \Psi = \overline{q}_n^{-1} \circ \prod_k \lambda_{k,n}$ for every $n \geq 0$. For continuity of Ψ: given n, if $(\hat{x}_k)_k \in \prod_k \ker \lambda_{k,n}$ then $\lambda_n\left(\Psi\left((\hat{x}_k)_k\right)\right) = \overline{q}_n^{-1}\left((\lambda_{k,n}(\hat{x}_k))_k\right) = \overline{q}_n^{-1}(0) = 0$, so $\Psi\left(\prod_k \ker \lambda_{k,n}\right) \subseteq \ker \lambda_n$ (same index n). For continuity of Ψ^{-1}: given n, if $x \in \ker \lambda_n$, applying $\overline{q}_n$ to $\lambda_n(x) = 0$ gives $0 = \overline{q}_n(\lambda_n(x)) = \left(\lambda_{k,n}(\Psi^{-1}(x)_k)\right)_k$ (the displayed equation, applied to $\Psi^{-1}(x)$ and rearranged), so $\lambda_{k,n}(\Psi^{-1}(x)_k) = 0$ for every k, that is $\Psi^{-1}(x) \in \prod_k \ker \lambda_{k,n}$: $\Psi^{-1}(\ker \lambda_n) \subseteq \prod_k \ker \lambda_{k,n}$ (same index n). ∎

Lemma 3.14.23 (Continuity into a Finite Product)
Let $(A, \mathfrak{a}_\bullet)$ be a filtered ring, $(N, \mathcal{N}_\bullet)$ and $\left(M_i, \mathcal{M}_{i,\bullet}\right)_{i=1}^n$ filtered A-modules (3.14.2), and equip $\prod_{i=1}^n M_i$ with the product filtration (3.14.22). Then an A-module homomorphism $\phi : N \to \prod_{i=1}^n M_i$ is continuous (3.14.7) (with $\rho := \mathrm{id}_A$) if and only if each $\phi_i := \pi_i \circ \phi : N \to M_i$ is continuous, where π_i is the i-th projection. (Taking $N := A$ and each M_i as filtered rings over themselves recovers the corresponding test for (3.14.5).)

Proof. Drafted by Claude Sonnet 5, reviewed by David Rufino.

If ϕ is continuous, then for each J there is L with $\phi(\mathcal{N}_L) \subseteq \prod_i \mathcal{M}_{i,J}$; applying π_i, $\phi_i(\mathcal{N}_L) \subseteq \mathcal{M}_{i,J}$, so each ϕ_i is continuous.

Conversely, suppose each ϕ_i is continuous. Fix J: for each $i = 1, \ldots, n$ there is m_i with $\phi_i\left(\mathcal{N}_{m_i}\right) \subseteq \mathcal{M}_{i,J}$; as there are finitely many i, $L := \max_i m_i$ satisfies $\mathcal{N}_L \subseteq \mathcal{N}_{m_i}$ for every i (the filtration being decreasing), so $\phi_i(\mathcal{N}_L) \subseteq \mathcal{M}_{i,J}$ for every i, that is $\phi(\mathcal{N}_L) \subseteq \prod_i \mathcal{M}_{i,J}$: ϕ is continuous. ∎

Lemma 3.14.24 (Projections from the Completion are Surjective)
Let $(M, \mathcal{M}_\bullet)$ be a filtered A-module, and $\widehat{M}$, λ_n as in (3.14.17). Then each λ_n is surjective. (In particular, this applies to $\hat{A}_\mathfrak{a}$, λ_n as in (3.14.19), the case $M := A$, $\mathfrak{a}_n := \mathfrak{a}^{n+1}$.)

Proof. Drafted by Claude Sonnet 5, reviewed by David Rufino.

Given $\overline{v}_n \in M/\mathcal{M}_n$, choose any representative $v_n \in M$. As each reduction $M/\mathcal{M}_{m+1} \twoheadrightarrow M/\mathcal{M}_m$ is surjective (3.14.17), recursively choose $v_{m+1} \in M$ reducing to (the class of) v_m for every $m \geq n$; for $m < n$, let $\overline{v}_m$ be the reduction of v_n. This gives a compatible sequence $(\overline{v}_m)_{m \geq 0} \in \widehat{M}$ with $\lambda_n\left((\overline{v}_m)_m\right) = \overline{v}_n$. ∎

Proposition 3.14.25 (Completion Matches the Filtration)
Let $(M, \mathcal{M}_\bullet)$ be a filtered A-module, with completion $\widehat{M}$, ι, λ_n (3.14.17). For every $n \geq 0$, $\iota(\mathcal{M}_n) \subseteq \ker \lambda_n \subseteq \ker \lambda_{n-1}$ (the latter for $n \geq 1$), and, writing $\overline{\pi}_n : M \to M/\mathcal{M}_n$ and $\pi_n : \widehat{M} \to \widehat{M}/\ker \lambda_n$ for the quotient maps, there is a unique A-module homomorphism $\nu_n : M/\mathcal{M}_n \to \widehat{M}/\ker \lambda_n$ (3.3.100) such that the square

$$\begin{array}{ccc} M & \xrightarrow{\ \iota\ } & \widehat{M} \\ {\scriptstyle \overline{\pi}_n}\downarrow & & \downarrow{\scriptstyle \pi_n} \\ M/\mathcal{M}_n & \xrightarrow[\nu_n]{} & \widehat{M}/\ker \lambda_n \end{array}$$

commutes, that is $\nu_n \circ \overline{\pi}_n = \pi_n \circ \iota$; and ν_n is an isomorphism, with inverse $\mu_n := \nu_n^{-1}$ satisfying $\mu_n \circ \pi_n = \lambda_n$. Moreover, writing $\rho_n : M/\mathcal{M}_n \to M/\mathcal{M}_{n-1}$ for the natural reduction (well-defined by (3.3.100), as $\mathcal{M}_n \subseteq \mathcal{M}_{n-1}$) and $\hat{\rho}_n : \widehat{M}/\ker \lambda_n \to \widehat{M}/\ker \lambda_{n-1}$ for the one induced by $\ker \lambda_n \subseteq \ker \lambda_{n-1}$, $(\nu_n)_{n \geq 0}$ is natural in n: for every $n \geq 1$, the square

$$\begin{array}{ccc} M/\mathcal{M}_n & \xrightarrow[\sim]{\nu_n} & \widehat{M}/\ker \lambda_n \\ {\scriptstyle \rho_n}\downarrow & & \downarrow{\scriptstyle \hat{\rho}_n} \\ M/\mathcal{M}_{n-1} & \xrightarrow[\sim]{\nu_{n-1}} & \widehat{M}/\ker \lambda_{n-1} \end{array}$$

commutes, that is $\hat{\rho}_n \circ \nu_n = \nu_{n-1} \circ \rho_n$ (equivalently, $\rho_n \circ \mu_n = \mu_{n-1} \circ \hat{\rho}_n$). (When $M := A$, ν_n, μ_n are ring isomorphisms, and moreover the stronger inclusion $\mathfrak{a}_n \hat{A} \subseteq \ker \lambda_n$ holds: for $c \in \mathfrak{a}_n$, $y \in \hat{A}$, $\lambda_n(\iota(c)y) = \lambda_n(\iota(c))\lambda_n(y) = 0$, using that $\hat{A}$ is then a ring and λ_n a ring homomorphism.)

© David Rufino CC BY-NC-ND 4.0

Proof. Drafted by Claude Sonnet 5, reviewed by David Rufino.

$\iota(\mathcal{M}_n) \subseteq \ker\lambda_n \subseteq \ker\lambda_{n-1}$. For $c \in \mathcal{M}_n$, $\lambda_n(\iota(c)) = c \bmod \mathcal{M}_n = 0$ by definition of ι, giving $\iota(\mathcal{M}_n) \subseteq \ker\lambda_n$. For $n \geq 1$: if $x = (\overline{v}_m)_m$ has $\lambda_n(x) = \overline{v}_n = 0$, then $\overline{v}_{n-1}$, being the reduction of $\overline{v}_n$ modulo $\mathcal{M}_{n-1}$ (as $(\overline{v}_m)_m$ is compatible under reduction), is also 0; so $\lambda_{n-1}(x) = 0$, giving $\ker\lambda_n \subseteq \ker\lambda_{n-1}$.

ν_n. As $\iota(\mathcal{M}_n) \subseteq \ker\lambda_n$ (above), the composite $\pi_n \circ \iota : M \to \widehat{M}/\ker\lambda_n$ kills $\mathcal{M}_n = \ker\overline{\pi}_n$, so (3.3.100) gives the unique A-module homomorphism $\nu_n : M/\mathcal{M}_n \to \widehat{M}/\ker\lambda_n$ with $\nu_n \circ \overline{\pi}_n = \pi_n \circ \iota$, that is $\nu_n(v \bmod \mathcal{M}_n) = \pi_n(\iota(v))$ for $v \in M$.

ν_n is injective: if $\nu_n(v \bmod \mathcal{M}_n) = 0$, that is $\iota(v) \in \ker\lambda_n$, then $0 = \lambda_n(\iota(v)) = v \bmod \mathcal{M}_n$ (by definition of ι).

ν_n is surjective, and $\pi_n = \nu_n \circ \lambda_n$: given $x \in \widehat{M}$, let $v \in M$ be any representative of $\lambda_n(x) \in M/\mathcal{M}_n$ ($\overline{\pi}_n$ being surjective); then $\lambda_n(\iota(v)) = v \bmod \mathcal{M}_n = \lambda_n(x)$, so $x - \iota(v) \in \ker\lambda_n$, giving $\pi_n(x) = \pi_n(\iota(v)) = \nu_n(v \bmod \mathcal{M}_n) = \nu_n(\lambda_n(x))$; in particular every $\pi_n(x)$ is of the form $\nu_n(-)$.

So ν_n is an isomorphism. **μ_n.** Set $\mu_n := \nu_n^{-1}$; then $\mu_n \circ \pi_n = \mu_n \circ \nu_n \circ \lambda_n = \lambda_n$ (using $\pi_n = \nu_n \circ \lambda_n$, just shown), as stated.

Naturality. $\hat{\rho}_n$ is well-defined since $\ker\lambda_n \subseteq \ker\lambda_{n-1}$, and $\hat{\rho}_n \circ \pi_n = \pi_{n-1}$, $\rho_n \circ \overline{\pi}_n = \overline{\pi}_{n-1}$ (the standard compatibility of quotient maps). For $v \in M$: $\hat{\rho}_n(\nu_n(\overline{\pi}_n(v))) = \hat{\rho}_n(\pi_n(\iota(v))) = \pi_{n-1}(\iota(v)) = \nu_{n-1}(\overline{\pi}_{n-1}(v)) = \nu_{n-1}(\rho_n(\overline{\pi}_n(v)))$; as $\overline{\pi}_n$ is surjective, this gives $\hat{\rho}_n \circ \nu_n = \nu_{n-1} \circ \rho_n$. Inverting (composing with $\mu_n = \nu_n^{-1}$ on the right and $\mu_{n-1} = \nu_{n-1}^{-1}$ on the left) gives the equivalent $\rho_n \circ \mu_n = \mu_{n-1} \circ \hat{\rho}_n$. ∎

Proposition 3.14.26 (The Completion is Complete)
Let $(M, \mathcal{M}_\bullet)$ be a filtered A-module, with completion $\widehat{M}$, ι, λ_n (3.14.17), filtered by $\ker\lambda_\bullet$. Then $\left(\widehat{M}, \ker\lambda_\bullet\right)$ is separated (3.14.2) and complete (3.14.20). (When $M := A$ is a filtered ring, $\left(\hat{A}, \ker\lambda_\bullet\right)$ is moreover again a filtered ring (3.14.16).)

Proof. Drafted by Claude Sonnet 5, reviewed by David Rufino.

Write $\pi_n : \widehat{M} \to \widehat{M}/\ker\lambda_n$ for the quotient maps and $\rho_n : M/\mathcal{M}_n \to M/\mathcal{M}_{n-1}$, $\hat{\rho}_n : \widehat{M}/\ker\lambda_n \to \widehat{M}/\ker\lambda_{n-1}$ for the reduction maps. By (3.14.25), there is an isomorphism $\mu_n : \widehat{M}/\ker\lambda_n \xrightarrow{\sim} M/\mathcal{M}_n$ with $\mu_n \circ \pi_n = \lambda_n$, natural in n: $\rho_n \circ \mu_n = \mu_{n-1} \circ \hat{\rho}_n$.

Separated. If $x = (\overline{v}_n)_{n\geq 0} \in \widehat{M}$ (a compatible sequence, $\overline{v}_n \in M/\mathcal{M}_n$) has $\lambda_n(x) = 0$ for every n, that is $\overline{v}_n = 0$ for every n, then $x = 0$; so $\bigcap_n \ker\lambda_n = 0$, that is $\left(\widehat{M}, \ker\lambda_\bullet\right)$ is separated (3.14.2).

Complete (item b)). By the above, (3.14.20) applies to $\left(\widehat{M}, \ker\lambda_\bullet\right)$; we verify its item b). Let $(\overline{x}_n)_{n\geq 0}$, $\overline{x}_n \in \widehat{M}/\ker\lambda_n$, be compatible under reduction (that is $\hat{\rho}_n(\overline{x}_n) = \overline{x}_{n-1}$ for $n \geq 1$). Writing $\overline{v}_n := \mu_n(\overline{x}_n) \in M/\mathcal{M}_n$: for $n \geq 1$, naturality gives $\rho_n(\overline{v}_n) = \rho_n(\mu_n(\overline{x}_n)) = \mu_{n-1}(\hat{\rho}_n(\overline{x}_n)) = \mu_{n-1}(\overline{x}_{n-1}) = \overline{v}_{n-1}$: the sequence $(\overline{v}_n)_n$ is compatible. So $(\overline{v}_n)_n \in \widehat{M}$ itself, with $\lambda_n(\overline{v}_\bullet) = \overline{v}_n$ for every n (its own n-th coordinate); then $\mu_n(\pi_n(\overline{v}_\bullet)) = \lambda_n(\overline{v}_\bullet) = \overline{v}_n = \mu_n(\overline{x}_n)$, and μ_n injective gives $\pi_n(\overline{v}_\bullet) = \overline{x}_n$: the compatible sequence $(\overline{x}_n)_n$ arises from $\overline{v}_\bullet \in \widehat{M}$. So $\left(\widehat{M}, \ker\lambda_\bullet\right)$ is complete. ∎

Proposition 3.14.27 (Functoriality of Completion)
With completions $\widehat{M}$, ι_M, $\lambda_\bullet^M$ and $\widehat{M'}$, $\iota_{M'}$, $\lambda_\bullet^{M'}$ (3.14.17), let $\psi : M \to M'$ be a continuous ρ-homomorphism (3.14.7). Then there is a unique continuous ρ-homomorphism $\hat{\psi} : \widehat{M} \to \widehat{M'}$ (for the filtrations $\ker\lambda_\bullet^M$, $\ker\lambda_\bullet^{M'}$) with $\hat{\psi} \circ \iota_M = \iota_{M'} \circ \psi$.

Moreover this is functorial: $\widehat{\mathrm{id}_M} = \mathrm{id}_{\widehat{M}}$ (taking $\rho := \mathrm{id}_A$); and, for a further filtered A''-module $(N, \mathcal{N}_\bullet)$, ring homomorphism $\rho' : A' \to A''$, and continuous ρ'-homomorphism $\chi : M' \to N$ (so $\chi \circ \psi$ is a continuous $(\rho' \circ \rho)$-homomorphism), $\widehat{\chi \circ \psi} = \hat{\chi} \circ \hat{\psi}$.

*Moreover (**surjectivity criterion**), if ψ is taut (3.14.11) then $\hat{\psi}$ is surjective.*

*Moreover (**injectivity criterion**), if $\psi(\mathcal{M}_n) \subseteq \mathcal{M}'_n$ for every $n \geq 0$ and the induced map $\overline{\psi}_{n_J} : M/\mathcal{M}_{n_J} \to M'/\mathcal{M}'_{n_J}$ (3.3.100) is injective for every $J \geq 0$, for some strictly increasing sequence $n_0 < n_1 < n_2 < \cdots$, then $\hat{\psi}$ is injective.*

(Taking $M := A$, $M' := A'$ as filtered A-, A'-modules over themselves (3.14.3) and $\psi := \rho$ recovers (3.14.5) and the ring-level statement of this proposition exactly.)

Proof. Drafted by Claude Sonnet 5, reviewed by David Rufino.

Write $\overline{\pi}_n^M : M \to M/\mathcal{M}_n$, $\overline{\pi}_m^{M'} : M' \to M'/\mathcal{M}'_m$ for the canonical surjections, and $\rho^M_{n,n'} : M/\mathcal{M}_n \to M/\mathcal{M}_{n'}$ ($n \geq n'$) for M's reduction maps. Throughout, regard $M'/\mathcal{M}'_m$ as an A-module via restriction of scalars along ρ ($a \cdot x := \rho(a) \cdot x$); a ρ-linear map out of M (or a quotient of M) is then exactly an A-linear map into $M'/\mathcal{M}'_m$ for this restricted structure, so (3.3.100) applies to it unchanged.

© David Rufino CC BY-NC-ND 4.0

A compatible family of maps into the levels of $\widehat{M'}$. For each $m \geq 0$, by continuity of ψ, choose $n(m) \geq 0$ with $\psi\big(\mathcal{M}_{n(m)}\big) \subseteq \mathcal{M}'_m$; (3.3.100) then gives a unique A-linear (that is, ρ-linear) map $\overline{\psi}_{n(m),m} : M/\mathcal{M}_{n(m)} \to M'/\mathcal{M}'_m$ with $\overline{\psi}_{n(m),m} \circ \overline{\pi}^M_{n(m)} = \overline{\pi}^{M'}_m \circ \psi$. Set $\psi_m := \overline{\psi}_{n(m),m} \circ \lambda^M_{n(m)} : \widehat{M} \to M'/\mathcal{M}'_m$.

These are compatible as m varies: fix m, write $\tau_m : M'/\mathcal{M}'_{m+1} \to M'/\mathcal{M}'_m$ for the reduction, and set $n := \max(n(m), n(m+1))$. As $\mathcal{M}_n \subseteq \mathcal{M}_{n(m)}, \mathcal{M}_{n(m+1)}$, (3.3.100) gives $\overline{\psi}_{n,m} : M/\mathcal{M}_n \to M'/\mathcal{M}'_m$, $\overline{\psi}_{n,m+1} : M/\mathcal{M}_n \to M'/\mathcal{M}'_{m+1}$ (the same defining property, through the finer level n); by uniqueness there, $\overline{\psi}_{n(m),m} \circ \rho^M_{n,n(m)} = \overline{\psi}_{n,m}$, $\overline{\psi}_{n(m+1),m+1} \circ \rho^M_{n,n(m+1)} = \overline{\psi}_{n,m+1}$, and $\tau_m \circ \overline{\psi}_{n,m+1} = \overline{\psi}_{n,m}$ (all being the map $M/\mathcal{M}_n \to M'/\mathcal{M}'_m$ induced by ψ). As $\lambda^M_{n(m)} = \rho^M_{n,n(m)} \circ \lambda^M_n$ and $\lambda^M_{n(m+1)} = \rho^M_{n,n(m+1)} \circ \lambda^M_n$ (compatibility of $\widehat{M}$'s own projections),

$$\tau_m \circ \psi_{m+1} = \tau_m \circ \overline{\psi}_{n(m+1),m+1} \circ \rho^M_{n,n(m+1)} \circ \lambda^M_n = \tau_m \circ \overline{\psi}_{n,m+1} \circ \lambda^M_n = \overline{\psi}_{n,m} \circ \lambda^M_n = \overline{\psi}_{n(m),m} \circ \rho^M_{n,n(m)} \circ \lambda^M_n = \psi_m.$$

Existence. By the universal property of $\widehat{M'} = \varprojlim_m M'/\mathcal{M}'_m$ as an inverse limit (2.6.129), the compatible family $(\psi_m)_m$ gives a unique A-linear (that is, ρ-linear) map $\hat{\psi} : \widehat{M} \to \widehat{M'}$ with $\lambda^{M'}_m \circ \hat{\psi} = \psi_m$ for every m. (If $M := A$, $M' := A'$ and ψ is moreover a ring homomorphism with $\psi(\mathfrak{a}_n) \subseteq \mathfrak{a}'_{n(m)}$ at the same witnessing indices, $\hat{\psi}$ is a ring homomorphism too, by the same construction run with $\overline{\psi}_{n(m),m}$ the induced ring homomorphisms.)

For $v \in M$: $\lambda^{M'}_m\big(\hat{\psi}(\iota_M(v))\big) = \psi_m(\iota_M(v)) = \overline{\psi}_{n(m),m}\big(v \bmod \mathcal{M}_{n(m)}\big) = \psi(v) \bmod \mathcal{M}'_m = \lambda^{M'}_m(\iota_{M'}(\psi(v)))$; as m was arbitrary, $\hat{\psi}(\iota_M(v)) = \iota_{M'}(\psi(v))$: the square commutes.

For continuity of $\hat{\psi}$: fix m; for $x \in \ker \lambda^M_{n(m)}$, $\lambda^{M'}_m\big(\hat{\psi}(x)\big) = \psi_m(x) = \overline{\psi}_{n(m),m}(0) = 0$, so $\hat{\psi}(x) \in \ker \lambda^{M'}_m$: taking $n := n(m)$ verifies $\hat{\psi}(\ker \lambda^M_n) \subseteq \ker \lambda^{M'}_m$.

Uniqueness. Suppose $\hat{\psi}_1, \hat{\psi}_2 : \widehat{M} \to \widehat{M'}$ are both continuous ρ-homomorphisms with $\hat{\psi}_i \circ \iota_M = \iota_{M'} \circ \psi$. Fix $x \in \widehat{M}$ and $m \geq 0$; choose n_i with $\hat{\psi}_i\big(\ker \lambda^M_{n_i}\big) \subseteq \ker \lambda^{M'}_m$ $(i = 1, 2)$, and set $n := \max(n_1, n_2)$, so $\ker \lambda^M_n \subseteq \ker \lambda^M_{n_i}$ (3.14.17) and $\hat{\psi}_i(\ker \lambda^M_n) \subseteq \ker \lambda^{M'}_m$ for both i. By surjectivity of λ^M_n (3.14.24), choose $v \in M$ with $\lambda^M_n(\iota_M(v)) = \lambda^M_n(x)$, i.e. $\iota_M(v) - x \in \ker \lambda^M_n$. Then $\hat{\psi}_i(\iota_M(v)) - \hat{\psi}_i(x) = \hat{\psi}_i(\iota_M(v) - x) \in \ker \lambda^{M'}_m$, and $\hat{\psi}_i(\iota_M(v)) = \iota_{M'}(\psi(v))$; so $\hat{\psi}_i(x) \equiv \iota_{M'}(\psi(v)) \bmod \ker \lambda^{M'}_m$ for $i = 1, 2$, giving $\hat{\psi}_1(x) - \hat{\psi}_2(x) \in \ker \lambda^{M'}_m$. As $m \geq 0$ was arbitrary, $\hat{\psi}_1(x) - \hat{\psi}_2(x) \in \bigcap_m \ker \lambda^{M'}_m = 0$ (3.14.26) ($\widehat{M'}$ always separated); so $\hat{\psi}_1(x) = \hat{\psi}_2(x)$.

Identities. $\mathrm{id}_{\widehat{M}}$ is continuous and satisfies $\mathrm{id}_{\widehat{M}} \circ \iota_M = \iota_M \circ \mathrm{id}_M$; by the uniqueness just shown (applied to $\psi := \mathrm{id}_M$, $\rho := \mathrm{id}_A$), $\widehat{\mathrm{id}_M} = \mathrm{id}_{\widehat{M}}$.

Composition. $\hat{\chi} \circ \hat{\psi} : \widehat{M} \to \widehat{N}$ is continuous (a composite of continuous maps) and is a $(\rho' \circ \rho)$-homomorphism (a composite of ρ'- and ρ-linear maps), satisfying $\big(\hat{\chi} \circ \hat{\psi}\big) \circ \iota_M = \hat{\chi} \circ \iota_{M'} \circ \psi = \iota_N \circ \chi \circ \psi$; by uniqueness (applied to $\chi \circ \psi$), $\widehat{\chi \circ \psi} = \hat{\chi} \circ \hat{\psi}$.

Surjectivity criterion. Suppose ψ is taut (3.14.11), witnessed by a strictly increasing sequence $n_0 < n_1 < n_2 < \cdots$ with $M' = \psi(M) + \mathcal{M}'_{n_0}$ and $\mathcal{M}'_{n_J} = \psi\big(\mathcal{M}_{n_J}\big) + \mathcal{M}'_{n_{J+1}}$ for every $J \geq 0$; in particular $\psi(\mathcal{M}_n) \subseteq \mathcal{M}'_n$ for every $n \geq 0$, so, as above, $\overline{\psi}_n := \overline{\psi}_{n,n}$ is defined for every n, and $\hat{\psi} = \varprojlim \overline{\psi}$ by (3.3.155), taking $n(m) := m$ throughout. By (3.14.18) (applied to M and to M', with this sequence), $\widehat{M} \xrightarrow{\sim} \varprojlim_J M/\mathcal{M}_{n_J}$ and $\widehat{M'} \xrightarrow{\sim} \varprojlim_J M'/\mathcal{M}'_{n_J}$ (this identification of $\widehat{M'}$ with the compatible sequences of the subsequence needing no completeness of M', only its definition as an inverse limit), identifying $\hat{\psi}$ with $\varprojlim_J \overline{\psi}_{n_J}$; so it suffices to show this restricted map hits every compatible sequence $\overline{v}' = \big(\overline{v}'_{n_J}\big)_{J \geq 0}$, $\overline{v}'_{n_J} \in M'/\mathcal{M}'_{n_J}$. We build, together with representatives $v \in M$, a compatible sequence $\big(\overline{v}_{n_J}\big)_{J \geq 0}$, $\overline{v}_{n_J} \in M/\mathcal{M}_{n_J}$, with $\overline{\psi}_{n_J}\big(\overline{v}_{n_J}\big) = \overline{v}'_{n_J}$ for every J.

Choose any representative $v' \in M'$ of $\overline{v}'_{n_0}$. By $M' = \psi(M) + \mathcal{M}'_{n_0}$, write $v' = \psi(v) + d$ for some $v \in M$, $d \in \mathcal{M}'_{n_0}$; then $\psi(v) \equiv v' \bmod \mathcal{M}'_{n_0}$, that is $\overline{\psi}_{n_0}\big(\overline{v}_{n_0}\big) = \overline{v}'_{n_0}$ for $\overline{v}_{n_0} := \overline{\pi}^M_{n_0}(v)$. Inductively, given $v \in M$ with $\overline{\psi}_{n_J}\big(\overline{\pi}^M_{n_J}(v)\big) = \overline{v}'_{n_J}$ $(J \geq 0)$: choose a representative $c' \in M'$ of $\overline{v}'_{n_{J+1}}$; as $\psi(v)$ and c' have the same reduction mod $\mathcal{M}'_{n_J}$ (namely $\overline{\psi}_{n_J}\big(\overline{\pi}^M_{n_J}(v)\big) = \overline{v}'_{n_J}$ = the reduction of $\overline{v}'_{n_{J+1}}$, as $\overline{v}'$ is compatible), $\psi(v) - c' \in \mathcal{M}'_{n_J}$. By $\mathcal{M}'_{n_J} = \psi\big(\mathcal{M}_{n_J}\big) + \mathcal{M}'_{n_{J+1}}$, write $\psi(v) - c' = \psi(c) + e$ for some $c \in \mathcal{M}_{n_J}$, $e \in \mathcal{M}'_{n_{J+1}}$; then $\psi(v - c) - c' = -e \in \mathcal{M}'_{n_{J+1}}$, that is $\overline{\psi}_{n_{J+1}}\big(\overline{\pi}^M_{n_{J+1}}(v - c)\big) = \overline{v}'_{n_{J+1}}$. As $c \in \mathcal{M}_{n_J}$, $\overline{\pi}^M_{n_J}(v - c) = \overline{\pi}^M_{n_J}(v)$, so replacing v by $v - c$ preserves the sequence built so far (compatible at every earlier level) while extending it to level n_{J+1}; continuing the induction, and setting $\overline{v}_{n_J} := \overline{\pi}^M_{n_J}(v)$ at each stage for the (eventually fixed) representative v at that stage, gives the required compatible sequence.

Injectivity criterion. Suppose $\psi(\mathcal{M}_n) \subseteq \mathcal{M}'_n$ for every $n \geq 0$ and $\overline{\psi}_{n_J}$ is injective for every $J \geq 0$, for some strictly increasing sequence $n_0 < n_1 < n_2 < \cdots$. As above (taking $n(m) := m$ throughout), $\hat{\psi} = \varprojlim \overline{\psi}$ (3.3.155), and, by (3.14.18) (applied to M and to M', with this sequence), $\widehat{M} \xrightarrow{\sim} \varprojlim_J M/\mathcal{M}_{n_J}$ and $\widehat{M'} \xrightarrow{\sim} \varprojlim_J M'/\mathcal{M}'_{n_J}$, identifying $\hat{\psi}$ with $\varprojlim_J \overline{\psi}_{n_J}$. If $x = (\overline{v}_n)_{n \geq 0} \in \widehat{M}$ has $\hat{\psi}(x) = 0$, then (under this identification) $\overline{\psi}_{n_J}\big(\overline{v}_{n_J}\big) = 0$ for every J, so

© David Rufino CC BY-NC-ND 4.0

$\overline{v}_{n_J} = 0$ for every J ($\overline{\psi}_{n_J}$ injective); as the identification $\widehat{M} \xrightarrow{\sim} \varprojlim_J M/\mathcal{M}_{n_J}$, $x \mapsto \left(\overline{v}_{n_J}\right)_J$, is an isomorphism, $x = 0$. So $\hat{\psi}$ is injective. ∎

Proposition 3.14.28 (Universal Property of Completion)
With completion $\widehat{M}$, ι, λ_n (3.14.17), suppose moreover $(M', \mathcal{M}'_\bullet)$ is complete *(3.14.20), with $\iota_{M'} : M' \xrightarrow{\sim} \widehat{M}'$ its own completion, and let $\psi : M \to M'$ be a continuous ρ-homomorphism (3.14.7). This is (3.14.27), composed with $\iota_{M'}^{-1}$: there is a unique continuous ρ-homomorphism $\hat{\psi}$ (for the filtration $\ker \lambda_\bullet$ of $\widehat{M}$) making*

$$\begin{array}{ccc} M & \xrightarrow{\psi} & M' \\ {\scriptstyle\iota}\downarrow & \nearrow_{\hat{\psi}} & \\ \widehat{M} & & \end{array}$$

commute.

*Moreover (**surjectivity criterion**), if ψ is taut (3.14.11) then $\hat{\psi}$ is surjective.*

*Moreover (**injectivity criterion**), if $\psi(\mathcal{M}_n) \subseteq \mathcal{M}'_n$ for every $n \geq 0$ with $\overline{\psi}_{n_J} : M/\mathcal{M}_{n_J} \to M'/\mathcal{M}'_{n_J}$ injective for every $J \geq 0$, for some strictly increasing sequence $n_0 < n_1 < n_2 < \cdots$, then $\hat{\psi}$ is injective.*

(Taking $M := A$, $M' := A'$ as filtered A-, A'-modules over themselves and $\psi := \rho$ recovers the ring-level statement exactly.)

Proof. Drafted by Claude Sonnet 5, reviewed by David Rufino.

By (3.14.27), there is a unique continuous ρ-homomorphism $\Psi : \widehat{M} \to \widehat{M}'$ with $\Psi \circ \iota = \iota_{M'} \circ \psi$; set $\hat{\psi} := \iota_{M'}^{-1} \circ \Psi$, continuous as a composite, with $\hat{\psi} \circ \iota = \iota_{M'}^{-1} \circ \Psi \circ \iota = \iota_{M'}^{-1} \circ \iota_{M'} \circ \psi = \psi$: the triangle commutes.

Uniqueness. If $\hat{\psi}_1, \hat{\psi}_2 : \widehat{M} \to M'$ are both continuous ρ-homomorphisms with $\hat{\psi}_i \circ \iota = \psi$, then $\iota_{M'} \circ \hat{\psi}_i : \widehat{M} \to \widehat{M}'$ are both continuous with $\left(\iota_{M'} \circ \hat{\psi}_i\right) \circ \iota = \iota_{M'} \circ \psi$; by uniqueness in (3.14.27), $\iota_{M'} \circ \hat{\psi}_1 = \iota_{M'} \circ \hat{\psi}_2$, and as $\iota_{M'}$ is injective, $\hat{\psi}_1 = \hat{\psi}_2$.

Surjectivity criterion. If ψ is taut, (3.14.27)'s own surjectivity criterion gives Ψ surjective, hence $\hat{\psi} = \iota_{M'}^{-1} \circ \Psi$ surjective ($\iota_{M'}^{-1}$ an isomorphism).

Injectivity criterion. Under the stated hypotheses, (3.14.27)'s own injectivity criterion gives Ψ injective, hence $\hat{\psi} = \iota_{M'}^{-1} \circ \Psi$ injective. ∎

Remark 3.14.29
Density (3.14.8) is exactly density in the usual topological sense for the metric $d(x, y) := 2^{-\sup\{n : x - y \in \mathfrak{b}_n\}}$ on B (with $\sup \varnothing := -1$, so $d(x, y) = 0$ iff $x = y$): a ball of radius 2^{-n} about $b \in B$ is exactly $b + \mathfrak{b}_{n+1}$, so $\phi(A)$ meets every such ball iff $B = \phi(A) + \mathfrak{b}_{n+1}$ for every n, that is ϕ is dense.

Proposition 3.14.30 (Completion Matches the Filtration, $\mathfrak{a}$-adic Case)
Let A be a ring, $\mathfrak{a} \triangleleft A$ an ideal, and $\hat{A}_\mathfrak{a}$, $\iota_\mathfrak{a}$, with projections $\lambda_n : \hat{A}_\mathfrak{a} \to A/\mathfrak{a}^{n+1}$, as in (3.14.19). This is the case $\mathfrak{a}_n := \mathfrak{a}^{n+1}$ of (3.14.25): for every $n \geq 0$, $\mathfrak{a}^{n+1}\hat{A}_\mathfrak{a} \subseteq \ker \lambda_n \subseteq \ker \lambda_{n-1}$ (the latter for $n \geq 1$), there is a unique isomorphism $\nu_n : A/\mathfrak{a}^{n+1} \xrightarrow{\sim} \hat{A}_\mathfrak{a}/\ker \lambda_n$ making the analogous square commute, with inverse μ_n satisfying $\mu_n \circ \pi_n = \lambda_n$, and $(\nu_n)_{n \geq 0}$ is natural in n with respect to the reductions $A/\mathfrak{a}^{n+1} \to A/\mathfrak{a}^n$ and $\hat{A}_\mathfrak{a}/\ker \lambda_n \to \hat{A}_\mathfrak{a}/\ker \lambda_{n-1}$.

Corollary 3.14.31 (Adic Completion via any Exponent)
In the setting of (3.14.19), for every integer $e \geq 1$, there is a unique continuous ring homomorphism

$$\theta_{\mathfrak{a},e} : \hat{A} \xrightarrow{\sim} \varprojlim_N A/\mathfrak{a}^{Ne}$$

compatible with projections, given explicitly by

$$\theta_{\mathfrak{a},e}\left((\overline{a}_n)_{n \geq 0}\right) = (\overline{a}_{Ne-1})_{N \geq 1}.$$

which is furthermore a bicontinuous isomorphism.

Proof. Drafted by Claude Sonnet 5, reviewed by David Rufino.

This is (3.14.18) (for the $\mathfrak{a}$-adic filtration $\mathfrak{a}_n := \mathfrak{a}^{n+1}$ (3.14.4)), applied to the sequence $n_N := Ne - 1$ ($N \geq 1$, so $\mathfrak{a}_{n_N} = \mathfrak{a}^{Ne}$ exactly). ∎

© David Rufino CC BY-NC-ND 4.0

Remark 3.14.32
If A is a k-algebra, for some ring k, then each $A/\mathfrak{a}^{n+1}$ is again a k-algebra, and the tower of (3.14.19) is a tower in $\mathbf{Alg}_k$ (3.3.153) so $\hat{A}_{\mathfrak{a}}$ is itself a k-algebra, and $\iota_{\mathfrak{a}}$ (running the same construction in $\mathbf{Alg}_k$ rather than $\mathbf{Ring}$) is a k-algebra homomorphism. Likewise, in (3.14.31), each $A/\mathfrak{a}^{Ne}$ is a k-algebra and $\theta_{\mathfrak{a},e}$ is a k-algebra isomorphism.

3.14.1 Hensel's Lemma

Lemma 3.14.33 (Convergent Sums in a Complete Ring)
Let A be a ring and $\mathfrak{a} \triangleleft A$ an ideal such that A is $\mathfrak{a}$-adically complete (3.14.21). Let $(y_i)_{i\geq 0}$ be a sequence in A with $y_i \in \mathfrak{a}^i$ for every $i \geq 0$. Then there is a unique $s \in A$ – written $s =: \sum_{i\geq 0} y_i$ – with

$$s \equiv \sum_{i=0}^{N} y_i \mod \mathfrak{a}^{N+1} \qquad \text{for every } N \geq 0.$$

Proof. Drafted by Claude Sonnet 5, reviewed by David Rufino.

Set $s_N := \sum_{i=0}^{N} y_i \in A$ for $N \geq 0$. As $s_N - s_{N-1} = y_N \in \mathfrak{a}^N$ for $N \geq 1$, $(\overline{s}_N)_{N\geq 0}$, $\overline{s}_N := s_N \bmod \mathfrak{a}^{N+1}$, is compatible under reduction; by (3.14.21) item b) (A being $\mathfrak{a}$-adically complete), there is $s \in A$ with $s \equiv s_N \bmod \mathfrak{a}^{N+1}$ for every $N \geq 0$: existence. If s, s' both satisfy this, $s - s' \in \bigcap_N \mathfrak{a}^{N+1} = 0$ (separated): uniqueness. ∎

Lemma 3.14.34 (Units of a Complete Ring)
Let A be a ring and $\mathfrak{a} \triangleleft A$ an ideal such that A is $\mathfrak{a}$-adically complete (3.14.21). If $u \in A^\star$ and $x \in \mathfrak{a}$, then $u + x \in A^\star$.

Proof. Drafted by Claude Sonnet 5, reviewed by David Rufino.

As $u + x = u(1 + u^{-1}x)$ and $u^{-1}x \in \mathfrak{a}$ ($\mathfrak{a}$ an ideal), it suffices to treat $u = 1$: show $1 + y \in A^\star$ for $y \in \mathfrak{a}$.

As $(-y)^i \in \mathfrak{a}^i$ for every $i \geq 0$, (3.14.33) gives $s := \sum_{i\geq 0} (-y)^i \in A$ with $s \equiv s_N := \sum_{i=0}^{N} (-y)^i \bmod \mathfrak{a}^{N+1}$ for every $N \geq 0$.

For every $N \geq 0$: $(1+y)s_N = \sum_{i=0}^{N} (-y)^i + \sum_{i=0}^{N} y(-y)^i = \sum_{i=0}^{N} (-y)^i - \sum_{i=1}^{N+1} (-y)^i = 1 - (-y)^{N+1}$ (telescoping), and likewise $s_N(1+y) = 1 - (-y)^{N+1}$ (y commuting with its own powers); so

$$(1+y)s - 1 = (1+y)(s - s_N) + ((1+y)s_N - 1) = (1+y)(s - s_N) - (-y)^{N+1} \in \mathfrak{a}^{N+1}$$

($\mathfrak{a}^{N+1}$ an ideal, $s - s_N \in \mathfrak{a}^{N+1}$, $(-y)^{N+1} \in \mathfrak{a}^{N+1}$), and likewise $s(1+y) - 1 \in \mathfrak{a}^{N+1}$. As $N \geq 0$ was arbitrary, $(1+y)s - 1, s(1+y) - 1 \in \bigcap_N \mathfrak{a}^{N+1} = 0$ (separated): $1 + y \in A^\star$, with inverse s. ∎

Lemma 3.14.35 (Hensel's Lemma)
Let A be a ring and $\mathfrak{a} \triangleleft A$ an ideal such that A is $\mathfrak{a}$-adically complete (3.14.21). Let $f \in A[X]$, with formal derivative $f' \in A[X]$ (3.8.9) (writing $f(x)$, $f'(x)$ for their evaluations at $x \in A$ (3.8.5)). If $a_0 \in A$ has $f(a_0) \in \mathfrak{a}$ and $f'(a_0) \in A^\star$, then there is $a \in A$ with $f(a) = 0$ and $a \equiv a_0 \bmod \mathfrak{a}$.

Proof. Drafted by Claude Sonnet 5, reviewed by David Rufino.

Taylor expansion. By (3.8.10), $f(a+h) - f(a) - f'(a)h \in (h^2)$ for every $a, h \in A$. In particular (taking $a := y$, $h := x - y$ for $x, y \in A$), $f(x) - f(y) \in (x - y)$.

Construction. Define inductively $a_{n+1} := a_n - f'(a_n)^{-1} f(a_n)$ for $n \geq 0$ (well-defined once $f'(a_n) \in A^\star$, shown below).

Claim: $f(a_n) \in \mathfrak{a}^{2^n}$, $f'(a_n) \in A^\star$, and $a_{n+1} \equiv a_n \bmod \mathfrak{a}^{2^n}$, for every $n \geq 0$. Base case $n = 0$: given. Inductively, suppose $f(a_n) \in \mathfrak{a}^{2^n} =: \mathfrak{a}^m$ and $f'(a_n) \in A^\star$; write $u_n := f'(a_n)^{-1}$, $h_n := -u_n f(a_n) \in \mathfrak{a}^m$ ($\mathfrak{a}^m$ an ideal), so $a_{n+1} = a_n + h_n \equiv a_n \bmod \mathfrak{a}^m$. By the Taylor expansion,

$$f(a_{n+1}) \equiv f(a_n) + f'(a_n)h_n = f(a_n) - f'(a_n)u_n f(a_n) = f(a_n)(1 - f'(a_n)u_n) = 0 \mod (h_n^2)$$

(as $f'(a_n)u_n = 1$ exactly), so $f(a_{n+1}) \in (h_n^2) \subseteq \mathfrak{a}^{2m} = \mathfrak{a}^{2^{n+1}}$ ($h_n \in \mathfrak{a}^m$, $\mathfrak{a}^m$ an ideal). As $a_{n+1} \equiv a_n \bmod \mathfrak{a}^m \subseteq \mathfrak{a}$ (as $m = 2^n \geq 1$), and (telescoping) $a_n \equiv a_0 \bmod \mathfrak{a}$ for every n, the Taylor expansion's consequence $f'(a_{n+1}) - f'(a_0) \in (a_{n+1} - a_0) \subseteq \mathfrak{a}$ gives, by (3.14.34) ($u := f'(a_0) \in A^\star$, $x := f'(a_{n+1}) - f'(a_0) \in \mathfrak{a}$), $f'(a_{n+1}) = u + x \in A^\star$. This completes the induction.

Existence of a. The sequence $n_N := 2^N - 1$ is strictly increasing, and the Claim gives $a_N \equiv a_{N-1} \bmod \mathfrak{a}^{2^{N-1}} = \mathfrak{a}^{n_{N-1}+1}$ for every $N \geq 1$, so $(\overline{a}_N)_{N\geq 0}$, $\overline{a}_N := a_N \bmod \mathfrak{a}^{n_N+1}$, is compatible under reduction; by (3.14.21) item c), there is $a \in A$ with $a \equiv a_N \bmod \mathfrak{a}^{2^N}$ for every $N \geq 0$.

© David Rufino CC BY-NC-ND 4.0

a is the desired root. For every $N \geq 0$: $a - a_N \in \mathfrak{a}^{2^N}$, so, by the Taylor expansion's consequence, $f(a) - f(a_N) \in (a - a_N) \subseteq \mathfrak{a}^{2^N}$; as also $f(a_N) \in \mathfrak{a}^{2^N}$ (the Claim), $f(a) \in \mathfrak{a}^{2^N}$, for every $N \geq 0$. So $f(a) \in \bigcap_N \mathfrak{a}^{2^N} = \bigcap_n \mathfrak{a}^n = 0$ (separated), giving $f(a) = 0$; and, at $N = 0$, $a \equiv a_0 \bmod \mathfrak{a}^{2^0} = \mathfrak{a}$. ■

Corollary 3.14.36 (Idempotent Lifting)
Let A be a ring and $\mathfrak{a} \triangleleft A$ an ideal such that A is $\mathfrak{a}$-adically complete (3.14.21). Then every idempotent $\overline{e} \in A/\mathfrak{a}$ lifts to an idempotent $e \in A$, that is $e^2 = e$ and $e \equiv \overline{e} \bmod \mathfrak{a}$.

Proof. Drafted by Claude Sonnet 5, reviewed by David Rufino.

Pick $a_0 \in A$ representing $\overline{e}$, so $f(a_0) := a_0^2 - a_0 \in \mathfrak{a}$ for $f(X) := X^2 - X \in A[X]$, with $f'(X) = 2X - 1$ (3.8.9). As $\overline{e}^2 = \overline{e}$, $(2a_0 - 1)^2 = 4a_0^2 - 4a_0 + 1 = 4(a_0^2 - a_0) + 1 \equiv 1 \bmod \mathfrak{a}$ (writing $4(a_0^2 - a_0) = 4f(a_0) \in \mathfrak{a}$), so $2a_0 - 1 \in A^\star$ by (3.14.34) (applied with $u := 1$). So (3.14.35) gives $e \in A$ with $e^2 - e = 0$ and $e \equiv a_0 \equiv \overline{e} \bmod \mathfrak{a}$: the desired lift. ■

© David Rufino CC BY-NC-ND 4.0